ZOIKOSUITE

Data Model / ERD Pack

Governed Business Operations Intelligence Platform

Classification
CONFIDENTIAL — INTERNAL STRATEGIC DOCUMENT

Standard
Enterprise SaaS · Data Architecture Specification

Control Targets
ISO 27001 · SOC 2 Type II · GDPR · CCPA-aligned posture · jurisdiction-specific residency readiness

Architecture Style
Governance-First · Event-Driven · API-First · Multi-Entity · Multi-Jurisdiction

Version
1.1 — Sovereign Data Model / ERD Pack Refined

Document
04 of 06 in the ZoikoSuite Architecture Series

ARCHITECTURE SERIES

01 Sovereign Back-End Architecture
02 System Architecture Diagram Pack
03 Microservices Specification Pack
04 Data Model / ERD Pack *(this document)*
05 Security Architecture Specification
06 Engineering Build Blueprint

01 · PURPOSE OF THIS DOCUMENT

This document defines the **canonical truth model** of ZoikoSuite.

It specifies:

- authoritative business entities
- cross-domain relationship boundaries

- source-truth ownership rules
- key propagation across tenant, legal entity, jurisdiction, and residency scopes
- effective-dated and versioned modeling requirements
- audit and evidence lineage requirements
- immutable ledger and evidential record patterns
- ERD-level structure for core platform domains
- schema-governance constraints required to prevent architectural drift
- residency, sovereignty, and traceability rules required for regulator-grade operation

This document must be read with:

- **Document 01** — Sovereign Back-End Architecture
- **Document 02** — System Architecture Diagram Pack
- **Document 03** — Microservices Specification Pack

If Document 03 defines **which service owns which truth**, this document defines **what that truth must look like structurally**.

This is not a generic schema guide.
It is the **canonical structural specification of business truth** for ZoikoSuite.

02 · DATA MODELING DOCTRINE

ZoikoSuite data modeling is governed by the same doctrine as the platform itself.

2.1 Source Truth Must Be Singular

Every material object has one authoritative owning service and one authoritative write boundary.

2.2 Tenant, Entity, Jurisdiction, and Residency Are Mandatory Context Dimensions

Every material record must be explicitly traceable to:

- tenant
- legal entity
- jurisdictional applicability
- physical residency policy where relevant
- time of applicability

2.3 Effective Dating Is Not Optional

Tax rules, payroll rules, contracts, policies, delegated authority, organizational structures, residency assignments, and jurisdiction mappings change over time. Historical state must remain explainable against the rule state active at the time of execution.

2.4 No Silent Overwrite of Material History

Material business history must be append-only, versioned, adjustment-based, or tombstoned. Destructive overwrite is prohibited for critical objects.

2.5 Evidence Must Be Correlatable

Transactions, approvals, documents, events, decisions, workflows, and intelligence artifacts must be linkable through stable identifiers and correlation chains.

2.6 Derived Stores Are Not Authoritative

Warehouses, search indexes, caches, analytics projections, and benchmark datasets are derivative. Source domains remain authoritative.

2.7 Keys Must Be Stable and Portable

Every authoritative entity must have durable identifiers suitable for:

- event payloads
- integrations
- audit retrieval
- cross-domain joins
- evidence manifests
- external references where appropriate

2.8 Sensitive Data Must Be Classifiable at Schema Level

PII, payroll, banking, tax, legal, and privileged data must be classifiable at model level, not only in surrounding infrastructure.

2.9 Jurisdiction Is More Than Geography

Jurisdiction may exist at:

- country
- state or province
- tax authority boundary
- labor law boundary

- filing authority boundary
- document-residency boundary

The model must support this hierarchy explicitly.

2.10 Data Residency Is a Structural Constraint

The platform must know not only **which law applies**, but also **where the bytes are allowed to reside**.

This requires:

- explicit residency-region modeling
- residency policy assignment
- document and data-store placement awareness
- conflict handling where jurisdiction and residency obligations differ

2.11 Material Objects Do Not Soft-Delete

Critical objects must not disappear behind silent soft-delete flags. They must use:

- status transitions
- tombstone records
- archival state
- effective end-dating

This is essential for audit defensibility.

2.12 Schema Governance Is Mandatory

With many services and many teams, local schema drift will destroy platform coherence unless canonical structure is governed centrally.

ZoikoSuite therefore requires:

- centralized schema registry
- version-controlled schema definitions
- compatibility rules for event contracts
- controlled schema evolution discipline

03 · KEYING STRATEGY

ZoikoSuite requires a disciplined key strategy across services and stores.

3.1 Primary Identifiers

Every authoritative entity must have:

- internal UUID / ULID primary key
- stable public-safe reference where appropriate
- correlation support for events and evidence

3.2 Context Keys

Every material record must carry, directly or resolvably:

- `tenant_id`
- `legal_entity_id`
- `jurisdiction_id` or jurisdiction chain
- `data_residency_policy_id` where relevant
- `residency_region_id` where storage/location controls apply
- `created_at`
- `effective_from` / `effective_to` where applicable
- `created_by_principal_id` where human-caused
- `source_service`

3.3 Event Linkage Keys

For event and evidence correlation:

- `correlation_id`
- `causation_id`
- `workflow_instance_id`
- `governance_decision_id`
- `evidence_manifest_id`
- `document_version_id`
- `schema_version`

3.4 External Reference Keys

Where integrations exist:

- external provider ID
- bank reference
- filing authority reference
- tax authority reference
- HRIS source ID
- e-signature envelope ID

These never replace canonical internal identifiers.

3.5 Integrity Keys

Where evidential integrity matters, records may additionally carry:

- payload hash
- previous hash reference
- snapshot signature reference
- digital signature reference

These strengthen traceability and tamper detection.

04 · TOP-LEVEL ENTITY MAP

The ZoikoSuite data model is organized around these top-level canonical domains:

1. Tenant, Entity & Residency Core
2. Identity, Access & Authority
3. Policy, Jurisdiction & Governance
4. Finance
5. Payroll
6. HR & Workforce
7. Legal, Corporate & Contracts
8. Tax
9. Compliance & Obligations
10. Commercial Operations
11. Evidence, Events & Audit
12. Intelligence & Analytical Projection
13. Schema Governance & Interoperability

Each is specified below.

05 · TENANT, ENTITY & RESIDENCY CORE MODEL

This is the root of enterprise scope and sovereignty.

5.1 Core Entities

Tenant

Represents the customer organization boundary.

Core Attributes

- tenant_id
- tenant_code
- legal_name
- trading_name
- status
- default_currency_code
- primary_timezone
- primary_locale
- default_data_residency_policy_id
- created_at
- created_by
- lifecycle_state

LegalEntity

Represents a subsidiary, company, branch, or operational entity.

Core Attributes

- legal_entity_id
- tenant_id
- entity_code
- legal_name
- trading_name
- registration_number
- tax_registration_number
- tax_identity_bundle_id
- entity_type
- incorporation_date
- default_currency_code
- fiscal_calendar_id
- parent_legal_entity_id
- entity_status
- primary_jurisdiction_id
- data_residency_policy_id
- created_at

EntityHierarchy

Represents parent-child and reporting relationships.

Core Attributes

- hierarchy_id
- tenant_id
- parent_legal_entity_id
- child_legal_entity_id
- relationship_type

- effective_from
- effective_to

EntityJurisdictionAssignment

Represents the jurisdictions applicable to a legal entity.

Core Attributes

- assignment_id
- legal_entity_id
- jurisdiction_id
- assignment_type
- effective_from
- effective_to
- source_basis

DataResidencyPolicy

Defines where data for a tenant or entity is permitted to physically reside.

Core Attributes

- data_residency_policy_id
- tenant_id
- policy_name
- policy_code
- residency_mode
- conflict_resolution_mode
- active_flag

ResidencyRegion

Represents a physical hosting/storage region.

Core Attributes

- residency_region_id
- region_code
- region_name
- cloud_provider
- country_code
- sovereign_flag
- active_flag

TaxIdentityBundle

Versioned collection of tax registrations and evidence.

Core Attributes

- tax_identity_bundle_id
- legal_entity_id
- bundle_status
- effective_from
- effective_to
- primary_document_version_id nullable

UltimateBeneficialOwner

Represents the ownership chain up to beneficial owner level.

Core Attributes

- ubo_id
- tenant_id
- legal_entity_id
- owner_type
- owner_name
- ownership_percentage
- control_type
- sanctions_screening_status
- effective_from
- effective_to

FiscalCalendar

Represents reporting and close structure.

Core Attributes

- fiscal_calendar_id
- tenant_id
- calendar_name
- calendar_type
- start_month
- close_policy
- active_flag

5.2 ERD — Tenant, Entity & Residency Core

```
Tenant
 ├──< LegalEntity
 │      ├──< EntityJurisdictionAssignment >── Jurisdiction
 │      ├──< EntityHierarchy (parent/child self-reference)
 │      ├──> FiscalCalendar
 │      ├──> DataResidencyPolicy
```

```
│       ├──> TaxIdentityBundle
│       └──< UltimateBeneficialOwner
├──< FiscalCalendar
└──< DataResidencyPolicy >── ResidencyRegion
```

5.3 Modeling Rules

- every material execution object must resolve to one legal entity
- no `legal_entity_id` may exist without a `data_residency_policy_id`
- entity hierarchy is effective-dated
- jurisdiction assignment is effective-dated
- residency policy must be enforceable at platform and storage layers
- one entity may have multiple applicable jurisdictions simultaneously
- tax identity evidence must be document-linked and version-preserving
- UBO lineage must support compliance, KYC, and sanctions screening use cases

06 · IDENTITY, ACCESS & AUTHORITY MODEL

This model governs who may do what, where, and under what delegated authority.

6.1 Core Entities

Principal

- principal_id
- tenant_id
- principal_type
- identity_provider_subject
- email
- display_name
- status
- created_at

Role

- role_id
- tenant_id
- role_name
- role_code
- role_scope_type
- active_flag

PermissionBundle

- permission_bundle_id
- tenant_id
- bundle_name
- bundle_code
- active_flag

PrincipalRoleAssignment

- assignment_id
- principal_id
- role_id
- legal_entity_id nullable
- effective_from
- effective_to
- assigned_by

DelegatedAuthority

- delegated_authority_id
- delegator_principal_id
- delegate_principal_id
- scope_type
- legal_entity_id nullable
- authority_limit_type
- authority_limit_value
- effective_from
- effective_to
- revocation_status

SoDRule

- sod_rule_id
- tenant_id
- domain_code
- action_a
- action_b
- conflict_type
- jurisdiction_id nullable
- active_flag

AccessDecisionLog

- access_decision_log_id
- principal_id
- legal_entity_id
- action_type

- decision_outcome
- decision_basis
- correlation_id
- decided_at

6.2 ERD — Identity & Authority

```
Principal
 ├──< PrincipalRoleAssignment >── Role
 ├──< DelegatedAuthority (delegator)
 ├──< DelegatedAuthority (delegate)
 └──< AccessDecisionLog

Role
 └──< RolePermissionBundle >── PermissionBundle

Tenant
 ├──< Principal
 ├──< Role
 ├──< PermissionBundle
 └──< SoDRule
```

6.3 Modeling Rules

- access assignments must support entity scope
- delegated authority must be effective-dated
- SoD rules may be domain-specific and jurisdiction-specific
- access decisions are evidence and must not be ephemeral
- denials are as important as grants from an evidential standpoint

07 · POLICY, JURISDICTION & GOVERNANCE MODEL

This is the runtime rules foundation of ZoikoSuite.

7.1 Core Entities

Policy

- policy_id
- tenant_id
- policy_code
- policy_name

- policy_domain
- policy_status
- versioning_mode

PolicyVersion

- policy_version_id
- policy_id
- version_number
- effective_from
- effective_to
- policy_payload
- activation_status
- activated_by
- activated_at

Jurisdiction

- jurisdiction_id
- jurisdiction_code
- jurisdiction_name
- jurisdiction_type
- parent_jurisdiction_id
- authority_type
- active_flag

JurisdictionRule

- jurisdiction_rule_id
- jurisdiction_id
- rule_domain
- rule_code
- rule_name
- effective_from
- effective_to
- rule_payload
- source_reference
- rule_status
- external_feed_reference nullable
- legal_drift_state

GovernanceDecision

- governance_decision_id
- tenant_id
- legal_entity_id
- principal_id
- action_type

- action_subject_type
- action_subject_id
- policy_version_id nullable
- jurisdiction_rule_basis
- authorization_outcome
- workflow_instance_id nullable
- correlation_id
- decision_timestamp

EvidenceRequirement

- evidence_requirement_id
- tenant_id
- domain_code
- action_type
- evidence_type
- requirement_payload
- effective_from
- effective_to

TaxLogicSnapshot

Immutable point-in-time record of tax/rule basis used at execution.

Core Attributes

- tax_logic_snapshot_id
- legal_entity_id
- jurisdiction_id
- rule_source_type
- source_rule_id
- source_version
- snapshot_payload
- created_at
- correlation_id

7.2 ERD — Governance Model

```
Policy
 └──< PolicyVersion

Jurisdiction
 └──< JurisdictionRule
      └──< TaxLogicSnapshot

GovernanceDecision
 ├──> Tenant
 ├──> LegalEntity
```

```
├──> Principal
├──> PolicyVersion
└──> WorkflowInstance

EvidenceRequirement
 └── scoped by domain/action/effective date
```

7.3 Modeling Rules

- policies are versioned, never overwritten
- jurisdiction rules are effective-dated
- governance decisions must store basis, not just outcome
- tax/rule basis snapshots should be storable for downstream financial and payroll explainability
- legal drift indicators must not overwrite prior rule state

08 · FINANCE DOMAIN MODEL

Finance is a source-truth domain and must be modeled with strong consistency, audit discipline, and scalable read patterns.

8.1 Core Entities

Account

- account_id
- legal_entity_id
- chart_code
- account_code
- account_name
- account_type
- currency_mode
- active_flag

FiscalPeriod

- fiscal_period_id
- fiscal_calendar_id
- period_name
- period_start
- period_end
- close_status
- close_locked_at

JournalEntry

- journal_entry_id
- legal_entity_id
- fiscal_period_id
- journal_number
- journal_type
- posting_state
- source_event_id nullable
- governance_decision_id
- created_at
- posted_at
- posted_by

JournalLine

- journal_line_id
- journal_entry_id
- account_id
- line_number
- debit_amount
- credit_amount
- currency_code
- fx_rate
- tax_code nullable
- tax_logic_snapshot_id
- memo

Invoice

- invoice_id
- legal_entity_id
- counterparty_id
- invoice_type
- invoice_number
- invoice_date
- due_date
- invoice_status
- total_amount
- currency_code
- source_contract_id nullable

Payment

- payment_id
- legal_entity_id
- payment_type
- payment_status

- payment_reference
- bank_account_id
- counterparty_id
- amount
- currency_code
- requested_at
- settled_at

BankAccount

- bank_account_id
- legal_entity_id
- account_name
- masked_account_number
- bank_identifier
- currency_code
- account_status

ReconciliationRecord

- reconciliation_record_id
- legal_entity_id
- bank_account_id
- statement_date
- reconciliation_status
- matched_amount
- unmatched_amount

IntercompanyEntry

- intercompany_entry_id
- source_legal_entity_id
- target_legal_entity_id
- source_journal_entry_id
- target_journal_entry_id nullable
- match_status
- amount
- currency_code

BalanceSnapshot

Pre-computed signed balance view for performance and audit confidence.

Core Attributes

- balance_snapshot_id
- legal_entity_id
- fiscal_period_id

- snapshot_type
- snapshot_date
- account_id nullable
- balance_amount
- currency_code
- snapshot_signature
- generated_at

8.2 ERD — Finance

```
LegalEntity
 ├──< Account
 ├──< BankAccount
 ├──< Invoice
 ├──< Payment
 ├──< ReconciliationRecord
 ├──< BalanceSnapshot
 └──< JournalEntry
       ├──< JournalLine >── Account
       │     └──> TaxLogicSnapshot
       └──> FiscalPeriod

JournalEntry
 └──< IntercompanyEntry >── LegalEntity (target)

Invoice
 ├──> Counterparty
 └──> Contract (optional source)
```

8.3 Modeling Rules

- journal entry and line are append-safe
- posting states must support pending, validated, finalized
- every tax-sensitive posting should be traceable to a tax logic snapshot
- balance snapshots are derived but signed for integrity and performance efficiency
- derived consolidation must not overwrite base journal truth

09 · PAYROLL DOMAIN MODEL

Payroll must preserve calculation lineage, snapshot integrity, and mathematical explainability.

9.1 Core Entities

PayrollCycle

- payroll_cycle_id
- legal_entity_id
- payroll_frequency
- cycle_start
- cycle_end
- pay_date
- cycle_status

PayrollRun

- payroll_run_id
- legal_entity_id
- payroll_cycle_id
- run_number
- run_status
- snapshot_hash
- finalized_at
- governance_decision_id

PayrollResult

- payroll_result_id
- payroll_run_id
- employee_id
- gross_pay
- net_pay
- employer_cost
- currency_code
- result_status

PayrollTaxCalculation

- payroll_tax_calculation_id
- payroll_result_id
- jurisdiction_rule_id
- tax_engine_source
- tax_type
- taxable_amount
- tax_amount
- calculation_basis_payload

BenefitEnrollment

- benefit_enrollment_id
- employee_id
- benefit_plan_id

- enrollment_status
- effective_from
- effective_to

DeductionRecord

- deduction_record_id
- payroll_result_id
- deduction_type
- amount
- source_reference

GrossToNetCalculationLog

Explainability record for payroll math.

Core Attributes

- gross_to_net_log_id
- payroll_result_id
- calculation_sequence
- step_type
- step_label
- input_payload
- output_payload
- formula_expression
- engine_reference
- created_at

9.2 ERD — Payroll

```
LegalEntity
 └──< PayrollCycle
      └──< PayrollRun
           └──< PayrollResult >── Employee
                 ├──< PayrollTaxCalculation
                 ├──< DeductionRecord
                 └──< GrossToNetCalculationLog

Employee
 └──< BenefitEnrollment >── BenefitPlan
```

9.3 Modeling Rules

- payroll runs must be immutable after finalization
- payroll calculations must preserve jurisdiction and engine basis

- gross-to-net math must be explainable from stored records, not only code
- retroactive changes generate adjustments, not destructive rewrite
- payroll snapshots must be reproducible for regulator and audit review

10 · HR & WORKFORCE MODEL

This domain owns workforce core truth and lifecycle state.

10.1 Core Entities

Employee

- employee_id
- legal_entity_id
- employee_number
- principal_id nullable
- employment_status
- hire_date
- termination_date nullable
- worker_type
- primary_jurisdiction_id
- department_id
- position_id

EmploymentContract

- employment_contract_id
- employee_id
- contract_version_number
- contract_status
- contract_type
- effective_from
- effective_to
- compensation_reference
- signed_document_version_id nullable

Department

- department_id
- legal_entity_id
- department_code
- department_name
- parent_department_id nullable

Position

- position_id
- legal_entity_id
- position_code
- position_title
- job_family
- active_flag

LeaveRequest

- leave_request_id
- employee_id
- leave_type
- leave_start
- leave_end
- leave_status
- approver_principal_id

PerformanceReview

- performance_review_id
- employee_id
- review_cycle
- review_status
- reviewer_principal_id
- completed_at nullable

TerminationCase

- termination_case_id
- employee_id
- termination_reason
- notice_rule_basis
- case_status
- initiated_at
- effective_termination_date

10.2 ERD — HR & Workforce

```
LegalEntity
 ├──< Employee
 ├──< Department
 └──< Position

Employee
 ├──< EmploymentContract
 ├──< LeaveRequest
 ├──< PerformanceReview
 └──< TerminationCase
```

```
Employee
 ├──> Department
 └──> Position
```

10.3 Modeling Rules

- employee is authoritative workforce identity
- employment contracts are versioned
- termination must preserve legal basis and evidence linkage
- sensitive workforce fields require field-level classification and restricted access

11 · LEGAL, CORPORATE & CONTRACTS MODEL

This domain must support machine-trackable obligations and full lineage.

11.1 Core Entities

Contract

- contract_id
- legal_entity_id
- counterparty_id
- contract_number
- contract_status
- contract_type
- effective_from
- effective_to
- current_version_number
- governing_jurisdiction_id

ContractVersion

- contract_version_id
- contract_id
- version_number
- version_status
- drafted_at
- approved_at nullable
- executed_at nullable
- document_version_id

ContractClause

- contract_clause_id
- contract_version_id
- clause_code
- clause_type
- clause_text_hash
- clause_metadata

ContractObligation

- contract_obligation_id
- contract_id
- contract_version_id
- contract_clause_id
- obligation_type
- due_logic
- obligation_status
- responsible_role_code
- legal_entity_id

Resolution

- resolution_id
- legal_entity_id
- resolution_type
- resolution_number
- resolution_status
- effective_date
- document_version_id

CorporateAction

- corporate_action_id
- legal_entity_id
- action_type
- action_status
- filed_at nullable
- filing_reference nullable

Counterparty

- counterparty_id
- tenant_id
- counterparty_type
- legal_name
- registration_number
- risk_status

- validation_status

11.2 ERD — Legal & Contracts

```
LegalEntity
 ├──< Contract >── Counterparty
 │      └──< ContractVersion
 │             └──< ContractClause
 │
 ├──< Resolution
 └──< CorporateAction

Contract
 └──< ContractObligation
ContractVersion
 └──< ContractObligation
ContractClause
 └──< ContractObligation
```

11.3 Modeling Rules

- contracts must support version lineage
- obligations must be atomically linked to source clause and version
- legal objects must preserve signatory and approval evidence linkage
- no active contract state without approved governance path

12 · TAX MODEL

This domain must preserve effective-dated rule truth and explainability.

12.1 Core Entities

TaxRule

- tax_rule_id
- jurisdiction_id
- tax_rule_code
- tax_type
- effective_from
- effective_to
- calculation_payload
- source_reference
- status

TaxDetermination

- tax_determination_id
- legal_entity_id
- source_object_type
- source_object_id
- jurisdiction_id
- tax_rule_id
- determination_status
- taxable_basis
- calculated_tax_amount
- calculation_timestamp

TaxLiability

- tax_liability_id
- legal_entity_id
- tax_type
- filing_period_id
- jurisdiction_id
- accrued_amount
- settled_amount
- liability_status

TaxReturnDraft

- tax_return_draft_id
- legal_entity_id
- filing_period_id
- tax_type
- draft_status
- evidence_manifest_id nullable

TaxPayment

- tax_payment_id
- legal_entity_id
- tax_liability_id
- payment_status
- amount
- currency_code
- paid_at nullable

NexusRecord

- nexus_record_id
- legal_entity_id
- jurisdiction_id

- nexus_type
- effective_from
- effective_to
- source_basis

12.2 ERD — Tax

```
Jurisdiction
 └──< TaxRule

LegalEntity
 ├──< TaxDetermination
 ├──< TaxLiability
 ├──< TaxReturnDraft
 ├──< TaxPayment
 └──< NexusRecord

TaxLiability
 └──< TaxPayment
TaxRule
 └──< TaxDetermination
```

12.3 Modeling Rules

- tax rules are effective-dated
- determinations must retain exact rule basis
- liabilities and returns are entity-scoped and jurisdiction-scoped
- filing drafts should link to evidence sets where required

13 · COMPLIANCE & OBLIGATIONS MODEL

This is the governed accountability model.

13.1 Core Entities

Obligation

- obligation_id
- legal_entity_id
- jurisdiction_id
- obligation_source_type
- obligation_source_id
- obligation_code

- obligation_type
- obligation_status
- due_date
- severity_level
- responsible_function
- source_reference

FilingRequirement

- filing_requirement_id
- obligation_id
- filing_type
- filing_authority
- submission_channel
- filing_status

ComplianceStatus

- compliance_status_id
- legal_entity_id
- domain_code
- status_code
- status_reason
- evaluated_at
- evidence_sufficiency_state

ExceptionCase

- exception_case_id
- legal_entity_id
- exception_type
- severity_level
- linked_object_type
- linked_object_id
- case_status
- escalated_at nullable

EscalationRecord

- escalation_record_id
- exception_case_id
- escalated_to_role
- escalation_reason
- escalation_status
- escalated_at

13.2 ERD — Compliance

```
LegalEntity
 ├──< Obligation
 ├──< ComplianceStatus
 └──< ExceptionCase
       └──< EscalationRecord

Obligation
 └──< FilingRequirement
```

13.3 Modeling Rules

- obligation must always resolve to source origin
- obligation must be entity-bound and jurisdiction-bound
- compliance status must be explainable
- exceptions and escalations must preserve timeline integrity

14 · COMMERCIAL OPERATIONS MODEL

Commercial operations must align spend, approvals, and due diligence.

14.1 Core Entities

VendorProfile

- vendor_profile_id
- legal_entity_id
- counterparty_id
- vendor_status
- due_diligence_status
- risk_rating
- approved_at nullable

PurchaseRequest

- purchase_request_id
- legal_entity_id
- requester_principal_id
- request_status
- spend_category
- requested_amount
- currency_code
- justification_text

PurchaseOrder

- purchase_order_id
- legal_entity_id
- vendor_profile_id
- purchase_request_id nullable
- po_number
- po_status
- total_amount
- currency_code

InvoiceApproval

- invoice_approval_id
- invoice_id
- workflow_instance_id
- approval_status
- approved_at nullable

SpendThresholdConsumption

- spend_threshold_consumption_id
- legal_entity_id
- policy_version_id
- threshold_type
- consumed_amount
- period_reference

14.2 ERD — Commercial Ops

```
LegalEntity
 ├──< VendorProfile >── Counterparty
 ├──< PurchaseRequest
 ├──< PurchaseOrder >── VendorProfile
 └──< SpendThresholdConsumption

Invoice
 └──< InvoiceApproval >── WorkflowInstance
PurchaseRequest
 └──< PurchaseOrder
```

14.3 Modeling Rules

- commercial spend must resolve to policy and workflow evidence
- vendor state must link to due-diligence evidence
- invoice approval is workflow-backed, not a boolean-only field

15 · EVIDENCE, EVENTS & AUDIT MODEL

This is a core moat layer and must withstand Big Four and regulator scrutiny.

15.1 Core Entities

AuditEvent

- audit_event_id
- tenant_id
- legal_entity_id
- event_name
- event_version
- source_service
- actor_principal_id nullable
- correlation_id
- causation_id nullable
- event_timestamp
- payload_hash
- previous_event_hash nullable
- payload_reference

WorkflowInstance

- workflow_instance_id
- tenant_id
- legal_entity_id
- workflow_type
- workflow_status
- initiated_by
- started_at
- completed_at nullable

WorkflowTransition

- workflow_transition_id
- workflow_instance_id
- from_state
- to_state
- acted_by
- acted_at
- rationale nullable

Document

- document_id
- tenant_id
- legal_entity_id nullable
- document_type
- storage_uri
- retention_policy_code
- residency_policy_code
- created_at

DocumentVersion

- document_version_id
- document_id
- version_number
- checksum_hash
- uploaded_by
- uploaded_at
- supersedes_document_version_id nullable
- virus_scan_status
- digital_signature_id nullable

EvidenceManifest

- evidence_manifest_id
- tenant_id
- legal_entity_id
- manifest_type
- manifest_status
- generated_at
- generated_by
- scenario_reference

EvidenceManifestItem

- evidence_manifest_item_id
- evidence_manifest_id
- item_type
- item_reference_id
- source_service
- integrity_hash

15.2 ERD — Evidence & Audit

```
Tenant
 ├──< AuditEvent
 ├──< WorkflowInstance
 ├──< Document
 └──< EvidenceManifest
```

```
WorkflowInstance
 └──< WorkflowTransition

Document
 └──< DocumentVersion

EvidenceManifest
 └──< EvidenceManifestItem
```

15.3 Modeling Rules

- audit events are append-only
- workflow transitions are full-state history
- document versions must preserve lineage
- evidence manifests are structured, generated evidence sets
- evidential records must be retrieval-ready, not merely archived

15.4 Hash-Chain Integrity Rule

Every audit event should support a `previous_event_hash` reference to create a tamper-evident integrity chain.

This is not a marketing blockchain claim.
It is a regulator-grade integrity control.

15.5 Document Integrity Rule

Critical document versions should support:

- checksum validation
- malware / virus scanning status
- optional digital signature linkage
- residency-policy binding
- access lineage through evidential records

16 · INTELLIGENCE & ANALYTICAL PROJECTION MODEL

These are read-derived models and must not claim source truth.

16.1 Core Entities

AnomalyCase

- anomaly_case_id
- tenant_id
- legal_entity_id
- anomaly_type
- source_object_type
- source_object_id
- model_reference
- confidence_score
- anomaly_status
- detected_at

ForecastSnapshot

- forecast_snapshot_id
- tenant_id
- legal_entity_id
- forecast_type
- forecast_period
- model_reference
- generated_at
- scenario_label

ComplianceRiskScore

- compliance_risk_score_id
- legal_entity_id
- obligation_id nullable
- risk_score
- scoring_model_reference
- scored_at

ReconciliationSuggestion

- reconciliation_suggestion_id
- legal_entity_id
- source_record_a
- source_record_b
- confidence_score
- suggestion_status
- generated_at

16.2 Modeling Rules

- intelligence outputs are explainable artifacts
- intelligence records must preserve model reference and score basis

- intelligence does not mutate source truth
- acted-upon intelligence should remain evidentially linked to human outcomes

16.3 Commercial Projection Rule

Where legally and contractually permitted, anonymized and aggregated derived data may support:

- benchmarking products
- premium reporting
- peer-comparison analytics
- governed data-sharing tiers

This is derivative commercialization, never reuse of source truth in breach of tenant or privacy obligations.

17 · SCHEMA GOVERNANCE & INTEROPERABILITY MODEL

This is necessary to prevent service drift across a large platform.

17.1 Core Entities / Concepts

SchemaRegistryArtifact

- schema_registry_artifact_id
- schema_name
- schema_type
- version_number
- compatibility_mode
- owning_service
- registered_at
- active_flag

SchemaDependencyMap

- schema_dependency_map_id
- producer_service
- consumer_service
- schema_registry_artifact_id
- dependency_type

17.2 Modeling Rules

- all event schemas must be centrally registered
- compatibility mode must be declared
- breaking changes require controlled rollout
- local service convenience schemas may exist, but canonical event contracts must remain centrally governed
- schema evolution is an architecture discipline, not an ad hoc team decision

18 · CROSS-DOMAIN RELATIONSHIP RULES

The platform must preserve the following discipline:

18.1 Finance to Payroll

Payroll results may generate ledger impacts, but payroll does not own ledger truth.

18.2 HR to Payroll

Employee and contract truth influence payroll, but payroll results remain payroll-owned snapshots.

18.3 Contracts to Obligations

Contracts may generate obligations, but obligation lifecycle is governed in obligation models.

18.4 Tax to Finance / Payroll

Tax determinations derive from financial or payroll activity, but retain independent rule-basis records.

18.5 Workflow to Everything

Workflow may govern many actions, but does not own the domain object itself.

18.6 Evidence to Everything

Evidence links to all domains but does not replace domain truth.

18.7 Residency to Everything Sensitive

Residency policy is not isolated to storage architecture alone. Sensitive objects must be resolvable to residency policy at model level.

19 · EFFECTIVE-DATED MODELING RULES

The following objects must support effective dating or versioning:

- policy versions
- jurisdiction rules
- delegated authority
- entity hierarchy
- entity-jurisdiction assignment
- data residency assignment
- employment contracts
- compensation
- benefits eligibility
- tax rules
- obligation basis
- organizational structure where reporting changes matter
- tax identity bundles
- UBO structures

For all such objects:

- current state must be queryable
- historical state must be replayable
- future-dated state must be schedulable where business-valid

20 · DATA CLASSIFICATION MODEL

Each material field should be classifiable at schema design level.

20.1 Recommended Classification Tiers

- **Public** — safe for public publication
- **Internal** — non-public, low sensitivity
- **Confidential** — sensitive business data
- **Restricted** — PII, payroll, banking, tax, privilege-sensitive, or secrets-adjacent

20.2 Mandatory Restricted Categories

- national identifiers

- payroll amounts at individual level
- bank account references
- disciplinary records
- termination data
- legal draft artifacts where privileged
- tax identifiers
- credential or token references
- UBO identity data
- regulated compliance evidence where protected

Classification metadata should be accessible to access-control and residency-enforcement layers.

21 · AUTHORITATIVE KEY PROPAGATION RULES

Every material event, evidence record, and derived dataset must preserve:

- `tenant_id`
- `legal_entity_id`
- `jurisdiction_id` or resolvable jurisdiction chain
- `data_residency_policy_id` where relevant
- `source_service`
- `source_object_type`
- `source_object_id`
- `correlation_id`
- effective timestamp

Stripping context is architecturally prohibited.

22 · DATA INTEGRITY & CONSISTENCY RULES

22.1 Strong Consistency Required

For:

- journal posting
- payroll finalization
- authorization outcomes
- workflow transitions

- tenant/entity registry updates
- governance decision logging
- tax logic snapshot attachment to critical financial/payroll records

22.2 Eventual Consistency Acceptable

For:

- search indexes
- analytics projections
- intelligence projections
- benchmarking datasets
- notification delivery states

22.3 Compensating Transaction Discipline

For long-running distributed flows, corrections occur through explicit compensating actions, not hidden rewrite.

22.4 Tombstone Rule

Where a material object must be retired, archived, or logically removed, the model must prefer:

- state transitions
- tombstone records
- effective end-dating
  over soft-delete semantics that erase historical meaning.

23 · ERD PRIORITY PACK FOR ENGINEERING

The first ERDs engineering should implement are:

Phase A — The Root

- Tenant / LegalEntity / Jurisdiction / DataResidencyPolicy / ResidencyRegion
- Principal / Role / DelegatedAuthority
- Policy / PolicyVersion / JurisdictionRule
- GovernanceDecision / WorkflowInstance / AuditEvent

Phase B — The Movement

- JournalEntry / JournalLine / TaxLogicSnapshot / BalanceSnapshot
- PayrollResult / PayrollTaxCalculation / GrossToNetCalculationLog
- Contract / ContractVersion / ContractClause

Phase C — The Proof

- AuditEvent / Document / DocumentVersion
- EvidenceManifest / EvidenceManifestItem
- WorkflowTransition

Phase D — The Intelligence

- AnomalyCase
- ForecastSnapshot
- ComplianceRiskScore
- ReconciliationSuggestion

This sequencing protects the truth architecture before optimization layers proliferate.

24 · FINAL DATA MODEL DOCTRINE

ZoikoSuite data is not organized around screens, forms, or convenience tables.

It is organized around:

- truth ownership
- legal entity context
- jurisdictional applicability
- residency constraint
- governance dependency
- evidential consequence
- historical explainability
- structural interoperability

Every canonical object must answer six questions clearly:

1. Who owns this truth?
2. Which tenant and entity does it belong to?
3. Which jurisdictional and residency rules affect it?
4. How does its historical state remain explainable?
5. Which evidence and events must link to it?
6. Can it survive regulator, auditor, and forensic review without ambiguity?

If a data model cannot answer those questions clearly, it is not yet fit for ZoikoSuite.

CTO ASSESSMENT

This Data Model / ERD Pack brings ZoikoSuite to a genuine **distributed truth architecture standard** because it:

- translates service boundaries into authoritative truth models
- strengthens residency and sovereignty controls at model level
- preserves atomic traceability between action, rule, and evidence
- supports Big Four audit and regulator-grade explainability
- adds high-scale structures such as balance snapshots and calculation logs
- prevents schema drift through centralized contract governance
- creates a foundation for future monetization through governed analytical derivatives

This is where ZoikoSuite becomes not only service-oriented and truth-disciplined, but **forensically defensible**.